# Placard main board

Hierarchy: each box is one sheet; the sheets connect through global labels

## Power and clock

J13 USB-C, F201, U40 3.3 V

X1 25.175 MHz, U41 clock gate

HOLD / RUN jumpers

File: power.kicad_sch

## Control

U1 State, U2 Branch

U17 instruction register

File: control.kicad_sch

## Program counter and address mux

U3, U4 program counter

U7–U10 address mux

File: pc_mux.kicad_sch

## Memory and I/O decode

U31, U32 RAM, J8 cartridge

U11 memory control

U12 I/O decode, U25 map

File: memory.kicad_sch

## Video timing

U13 H, U14 V counters

U15 Sync, U35 audio slots

U29 line read

File: video_timing.kicad_sch

## ALU and registers

U5, U6 ALU

U19 AC, U20 X, U21 Y

File: alu_regs.kicad_sch

## Data bus

U18 D register

U26, U27 source drivers

R601–R608, RN601

File: data_bus.kicad_sch

## Expansion slots

J9–J12 card-edge slots

File: expansion.kicad_sch

## Video and audio out

U22 pixel buffer, U23 line code

U16 color decoder, VGA

U36 audio DAC, J6

File: video_out.kicad_sch

## Shared port

U37 latch, U38 port PLD

U39 rows, J1–J5

File: shared_port.kicad_sch

35 chips + U40 regulator (docs/CHIPS.md numbers; U24, U28, U30, U33, U34 are not used).
Decoupling: every chip's 100 nF Cn sits beside Un on its own sheet.
Power: +5V, GND, +3V3 (SD card), GNDA (audio, joined to GND on sheet 2).
The ROM cartridge is its own project: kicad/cartridge/.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

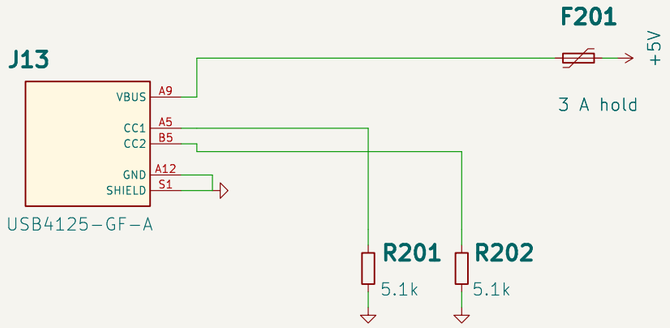
Sheet: /
File: placard.kicad_sch

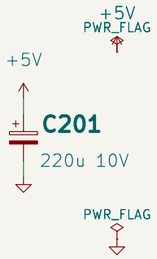
***Title: Placard: Main board***

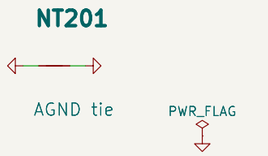
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/11 |

# Power and clock

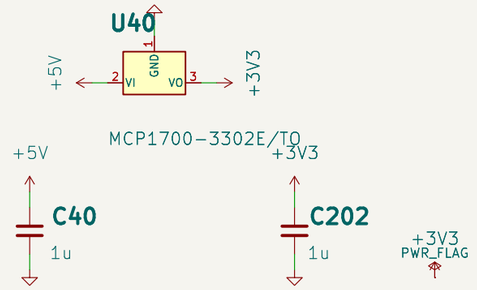
USB-C 5 V in, 3.3 V for the SD card, 25.175 MHz oscillator, clock gate CLK = OSC OR HOLD



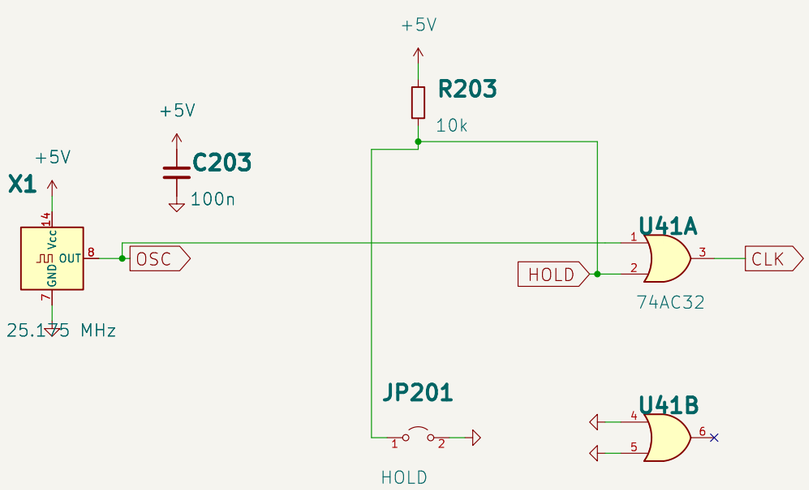



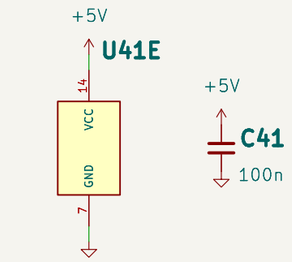

AGND (audio) meets GND here, at the power input

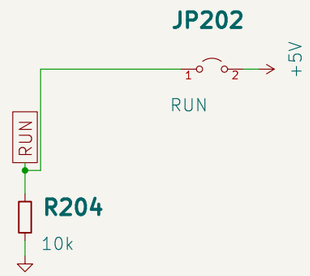



5.1k on CC1 and CC2: a USB-C source gives 5 V.
18 PLDs draw up to about 2.3 A (130 mA each at -7, worst case): use a 3 A supply; the polyfuse trips on a short.

U40: 3.3 V for the SD card only (sheet 10).
C40 input, C202 output (1 uF ceramic).




HOLD (10k up) and RUN (10k down) come from the cartridge. Without one, fit JP201 and JP202 for bring-up (BRINGUP.md); take them out with a cartridge.
Unused OR gates: inputs to GND.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

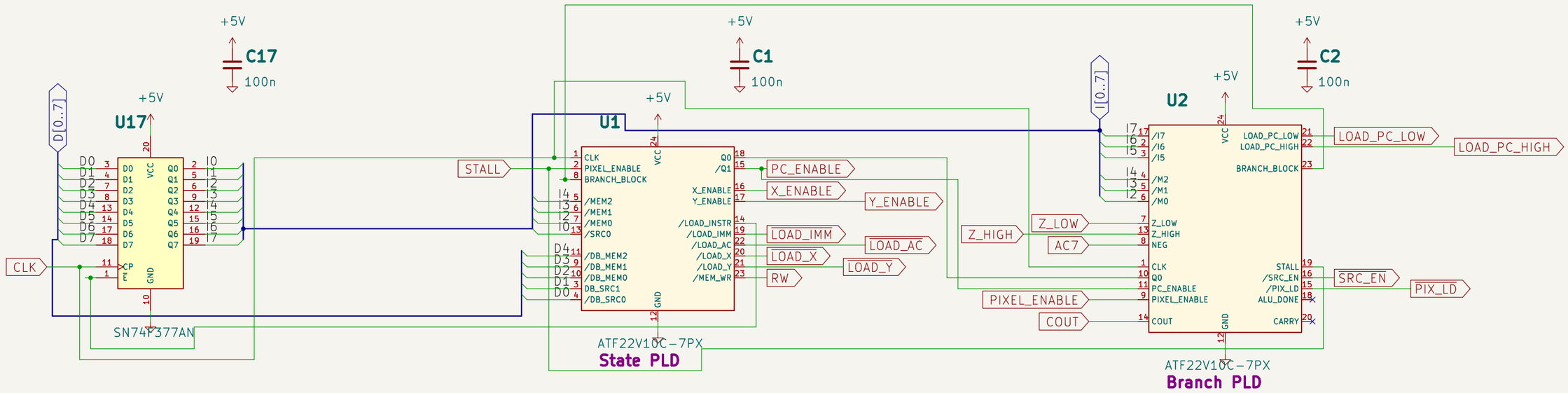
Sheet: /Power and clock/
File: power.kicad_sch

***Title: Placard: Power and clock***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/11 |

# Control

State and Branch PLDs, instruction register



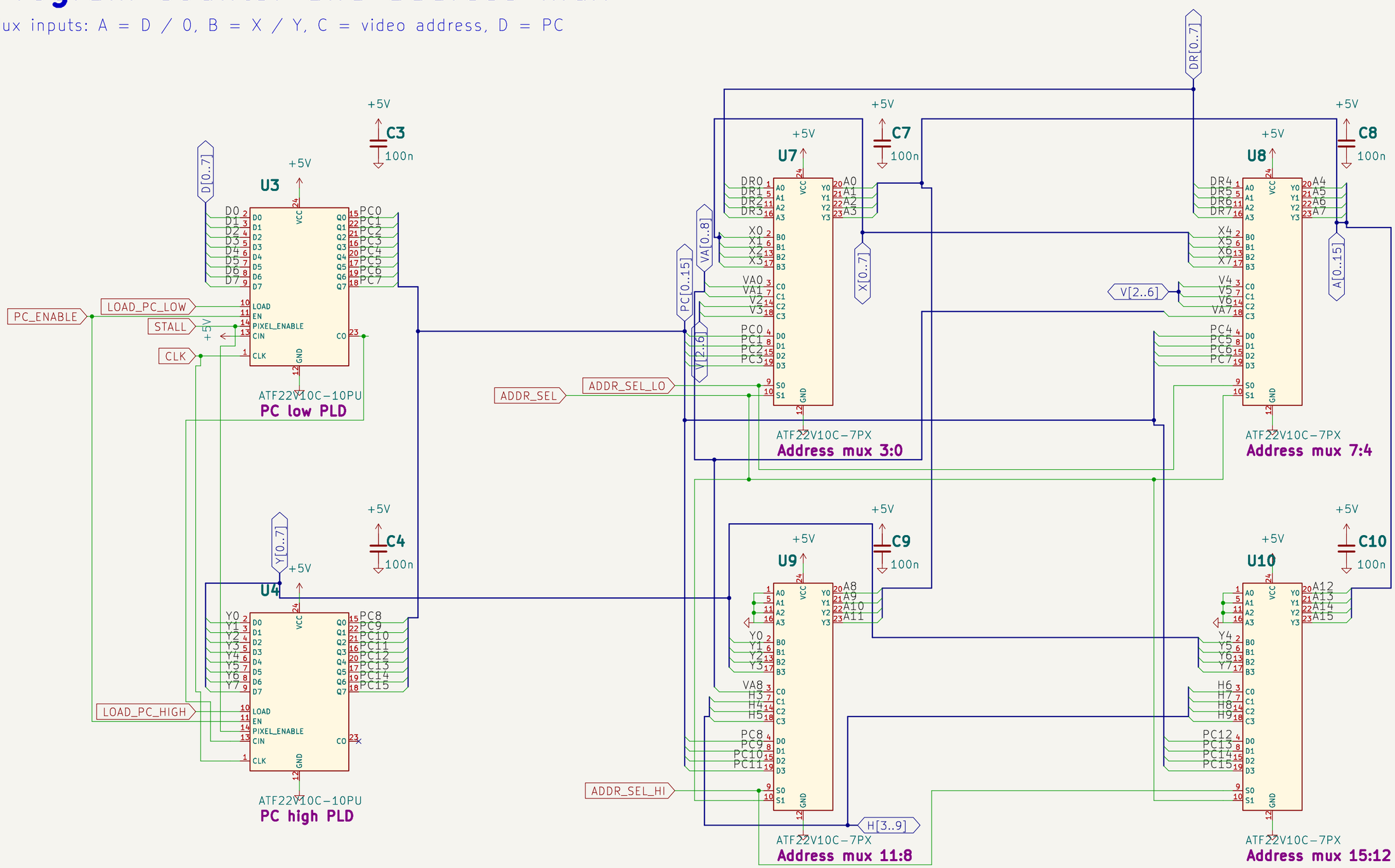

U17 loads the opcode in FETCH (/LOAD_INSTR). The byte is stored XOR FDh (CPU.md).
U1 State: FETCH / LOAD / EXEC, register loads (active low, F377 /E), RW (1 = read).
U2 Branch: NEG is AC7. ALU_DONE (18) and CARRY (20) are internal registers: no connection.
Bring-up straps S1-S3 go in U1's socket (BRINGUP.md), not on the board.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /Control/ | | |
| File: control.kicad_sch | | |
| ***Title: Placard: Control*** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/11 |

# Program counter and address mux

Mux inputs: A = D / 0, B = X / Y, C = video address, D = PC




Video address: VA0, VA1, VA7, VA8 from the audio slot PLD (line repeat), A6:A2 = V6:V2, A15:A9 = H9:H3.
U3 CIN tied high (always counts); U3 CO carries into U4. STALL holds the PC during video reads and the ALU wait.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

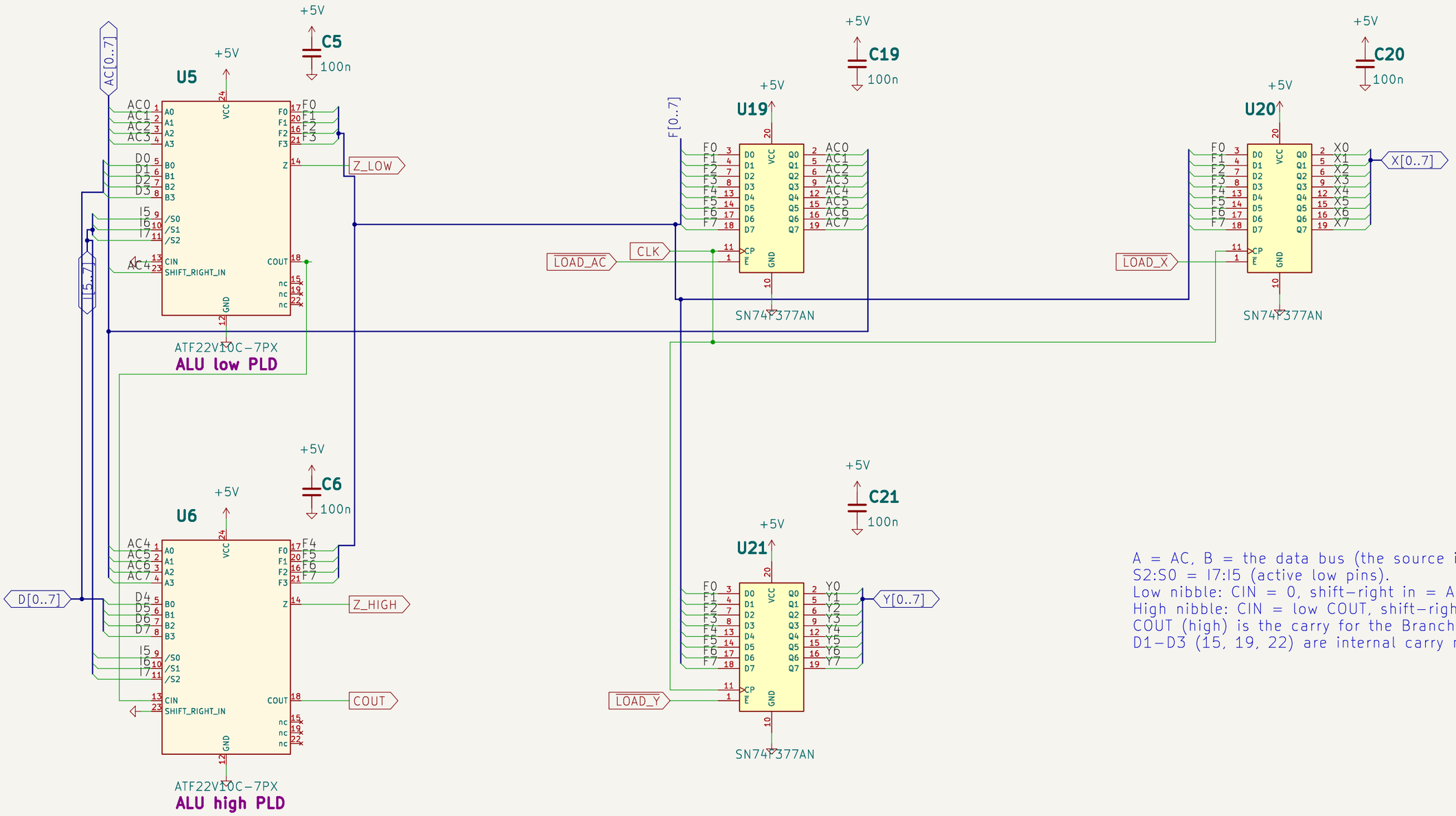
Sheet: /Program counter and address mux/
File: pc_mux.kicad_sch

***Title: Placard: Program counter and address mux***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/11 |

# ALU and registers

Two 4-bit ALU PLDs; F7:F0 loads AC, X or Y




A = AC, B = the data bus (the source in EXEC).
S2:S0 = I7:I5 (active low pins).
Low nibble: CIN = 0, shift-right in = AC4.
High nibble: CIN = low COUT, shift-right in = 0.
COUT (high) is the carry for the Branch PLD.
D1-D3 (15, 19, 22) are internal carry nodes.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

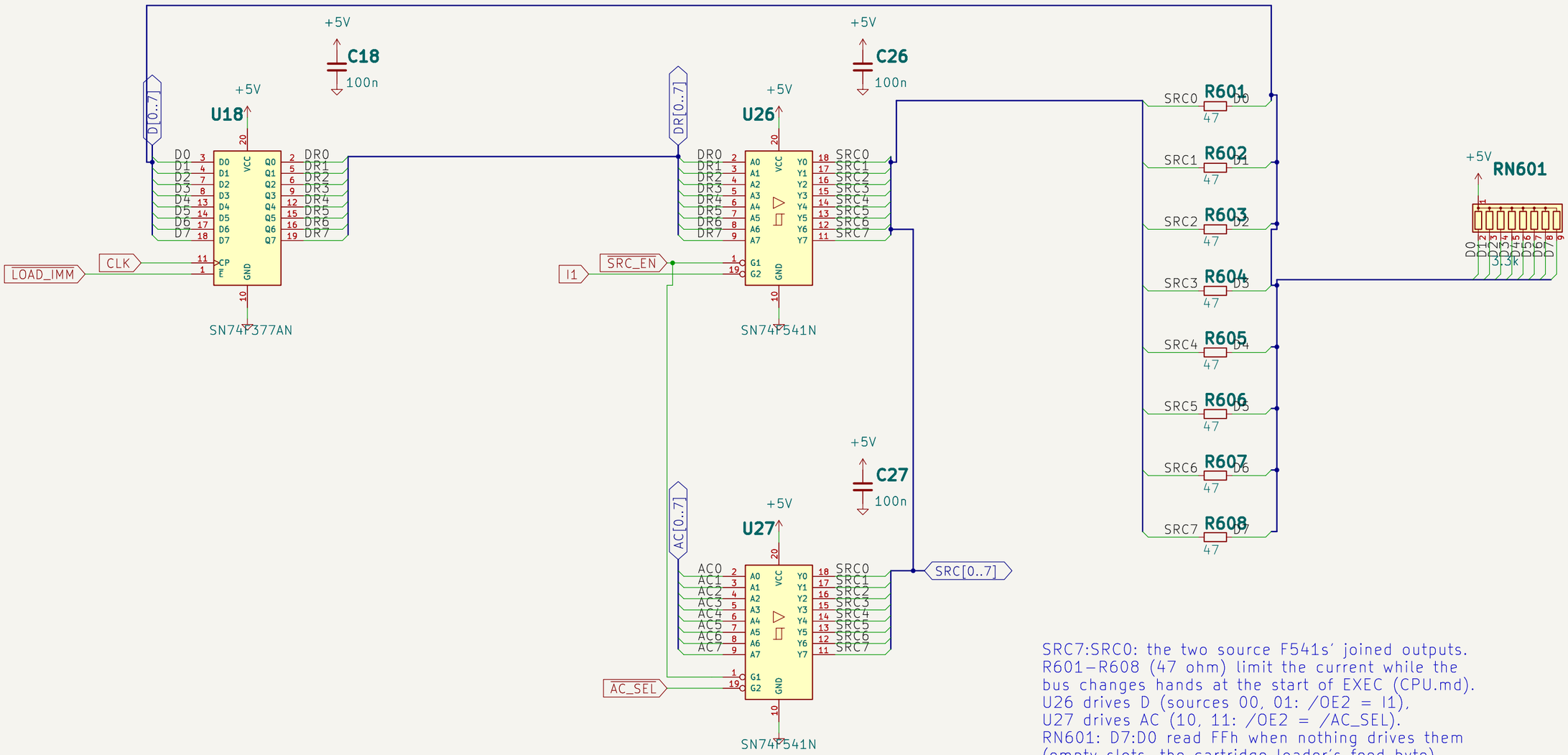
Sheet: /ALU and registers/
File: alu_regs.kicad_sch

***Title: Placard: ALU and registers***

| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/11 |

# Data bus

D register, source drivers (D or AC) onto the one data bus through series resistors, read pull-ups



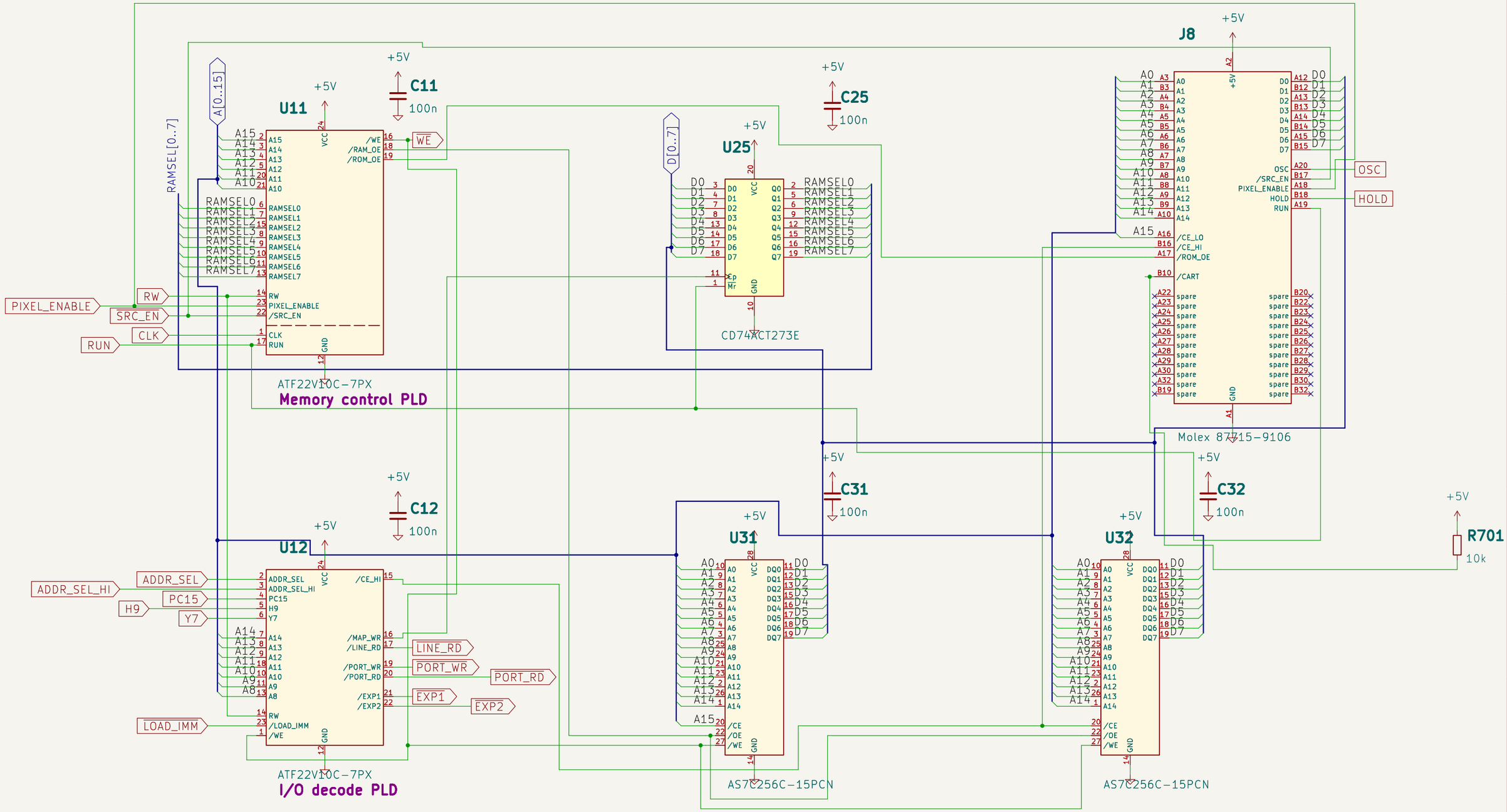

SRC7:SRC0: the two source F541s' joined outputs.
R601–R608 (47 ohm) limit the current while the
bus changes hands at the start of EXEC (CPU.md).
U26 drives D (sources 00, 01: /OE2 = I1),
U27 drives AC (10, 11: /OE2 = /AC_SEL).
RN601: D7:D0 read FFh when nothing drives them
(empty slots, the cartridge loader's feed byte).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | |
|---|---|
| Sheet: /Data bus/ | |
| File: data_bus.kicad_sch | |
| ***Title: Placard: Data bus*** | |
| Size: A3 | Date: 2026-10-20 |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | **Rev: 1** |
| | Id: 6/11 |

# Memory and I/O decode

RAM, ROM cartridge slot, memory control, I/O decode, map register




Lower 32K: /CE = A15 (RAM U31, cartridge /CE_LO); upper: /CE_HI from U12.
RUN low (cartridge loading): /ROM_OE held off, map register clear (all ROM).
/CART reads low with a cartridge in; nothing reads it yet.  22 spare contacts.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

**Placard computer**

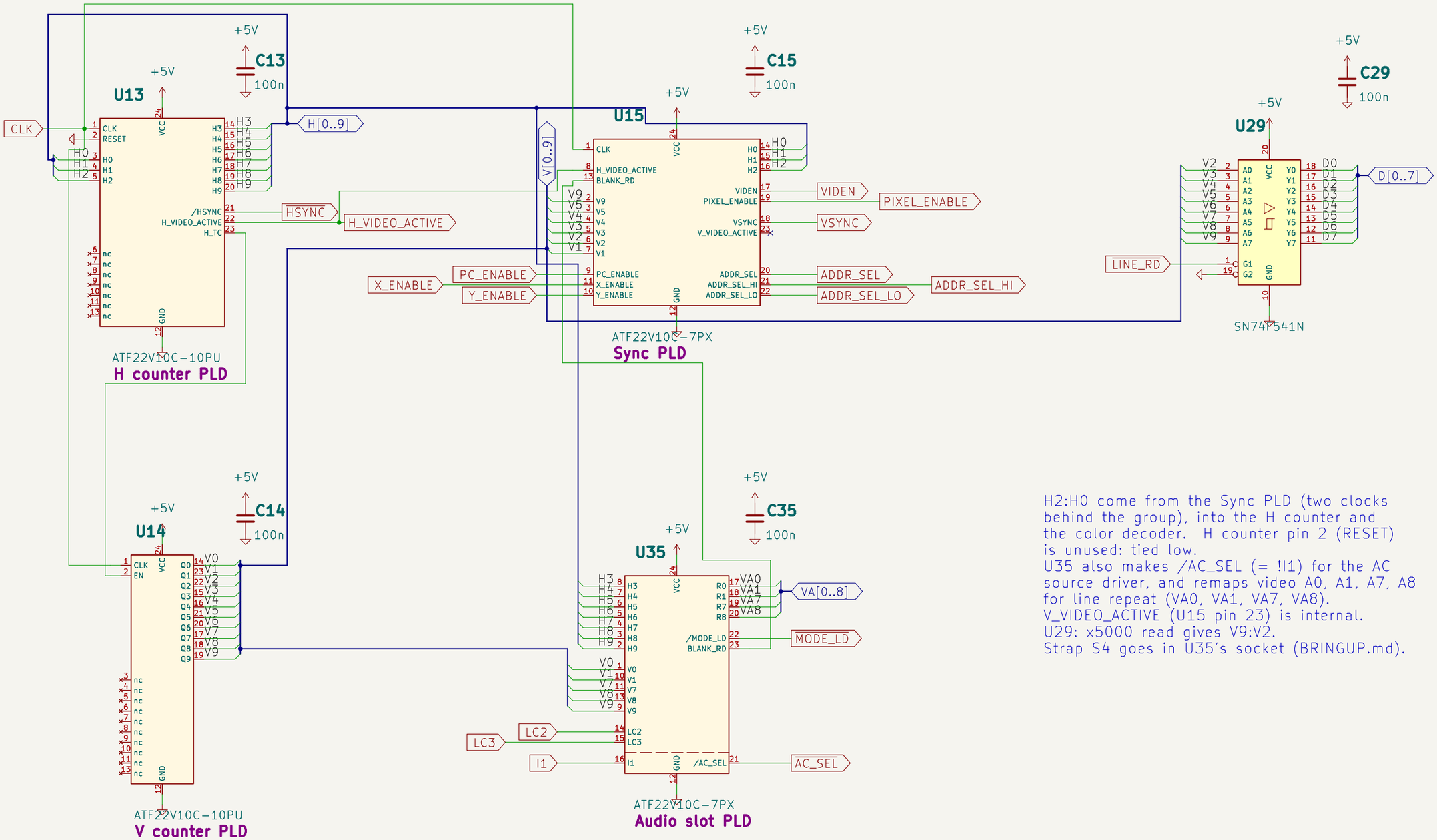
Sheet: /Memory and I/O decode/
File: memory.kicad_sch

***Title: Placard: Memory and I/O decode***

| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 7/11 |

# Video timing

640 x 480 at 60 Hz: H and V counters, Sync, mode and audio slots, line read



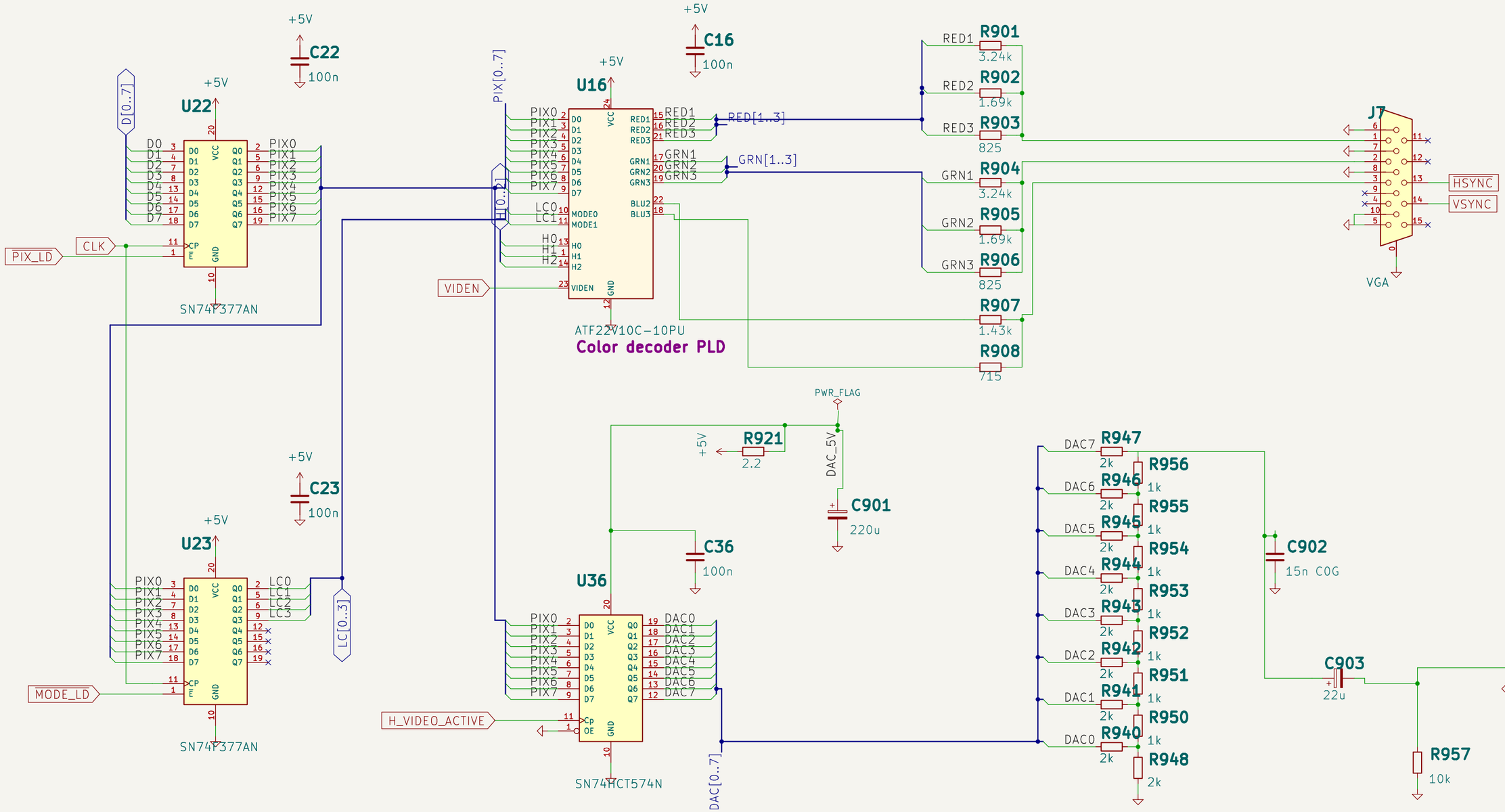

H2:H0 come from the Sync PLD (two clocks behind the group), into the H counter and the color decoder. H counter pin 2 (RESET) is unused: tied low.
U35 also makes /AC_SEL (= !I1) for the AC source driver, and remaps video A0, A1, A7, A8 for line repeat (VA0, VA1, VA7, VA8).
V_VIDEO_ACTIVE (U15 pin 23) is internal.
U29: x5000 read gives V9:V2.
Strap S4 goes in U35's socket (BRINGUP.md).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | | |
|---|---|---|
| Sheet: /Video timing/ | | |
| File: video_timing.kicad_sch | | |
| ***Title: Placard: Video timing*** | | |
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 8/11 |

# Video and audio out

Pixel buffer, line code, color decoder and VGA DAC; audio DAC register, R–2R ladder, headphones




U23 Q3:Q2 = line repeat to U35 (sheet 8); Q1:Q0 = color mode. DAC_5V: U36's own filtered rail (2.2 ohm, 220u, C36).
Ladder: 2k 1 % per bit + 2k termination, 1k between nodes (LAD7 = out). AGND (GNDA) joins GND at the power input.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

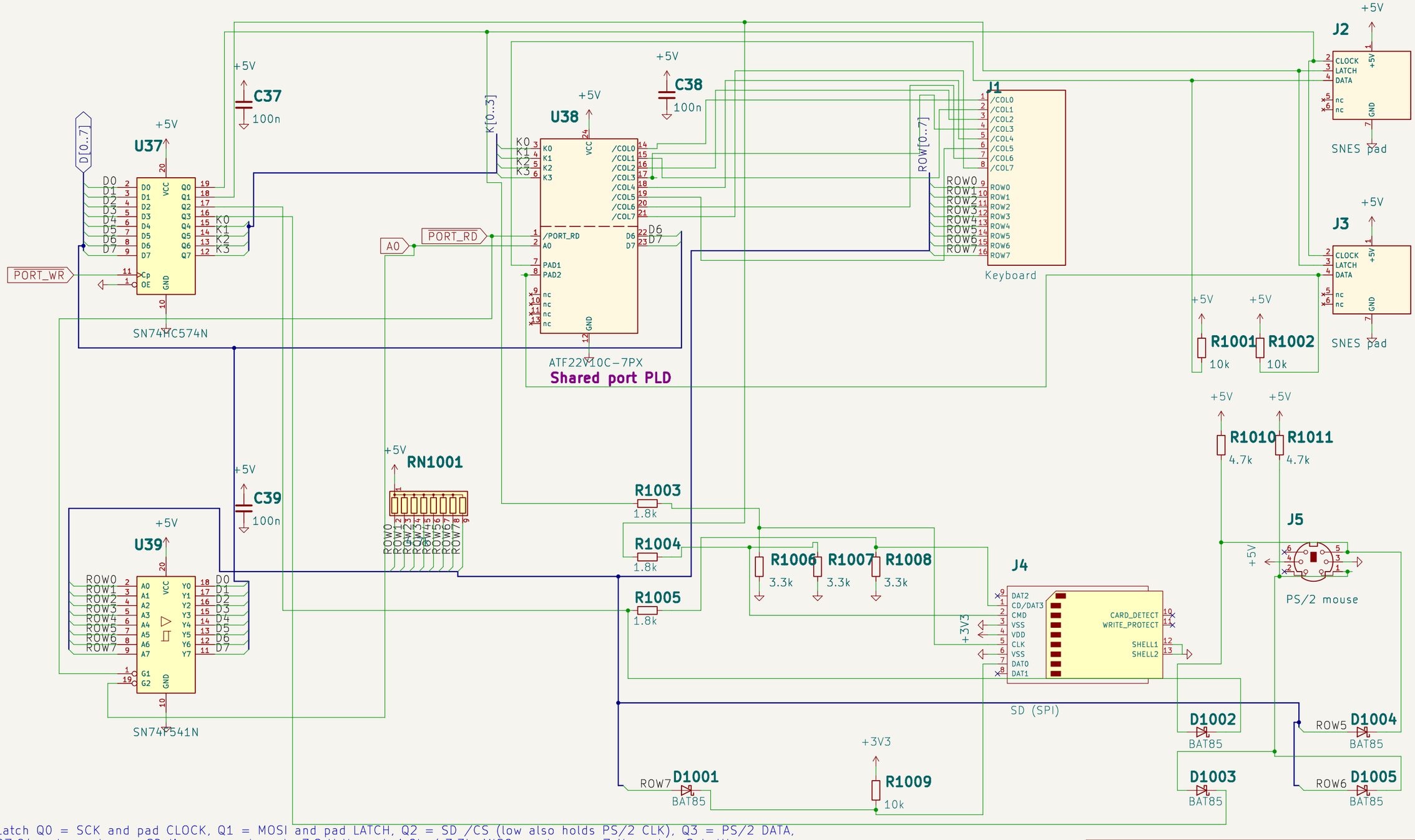
Sheet: /Video and audio out/
File: video_out.kicad_sch

***Title: Placard: Video and audio out***

| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 9/11 |

# Shared port x5200

Keyboard, two SNES pads, SD card (SPI), PS/2 mouse




Latch Q0 = SCK and pad CLOCK, Q1 = MOSI and pad LATCH, Q2 = SD /CS (low also holds PS/2 CLK), Q3 = PS/2 DATA, Q7:Q4 = key column. SD lines come down to 3.2 V through 1.8k / 3.3k; MISO reaches row 7 through a Schottky.

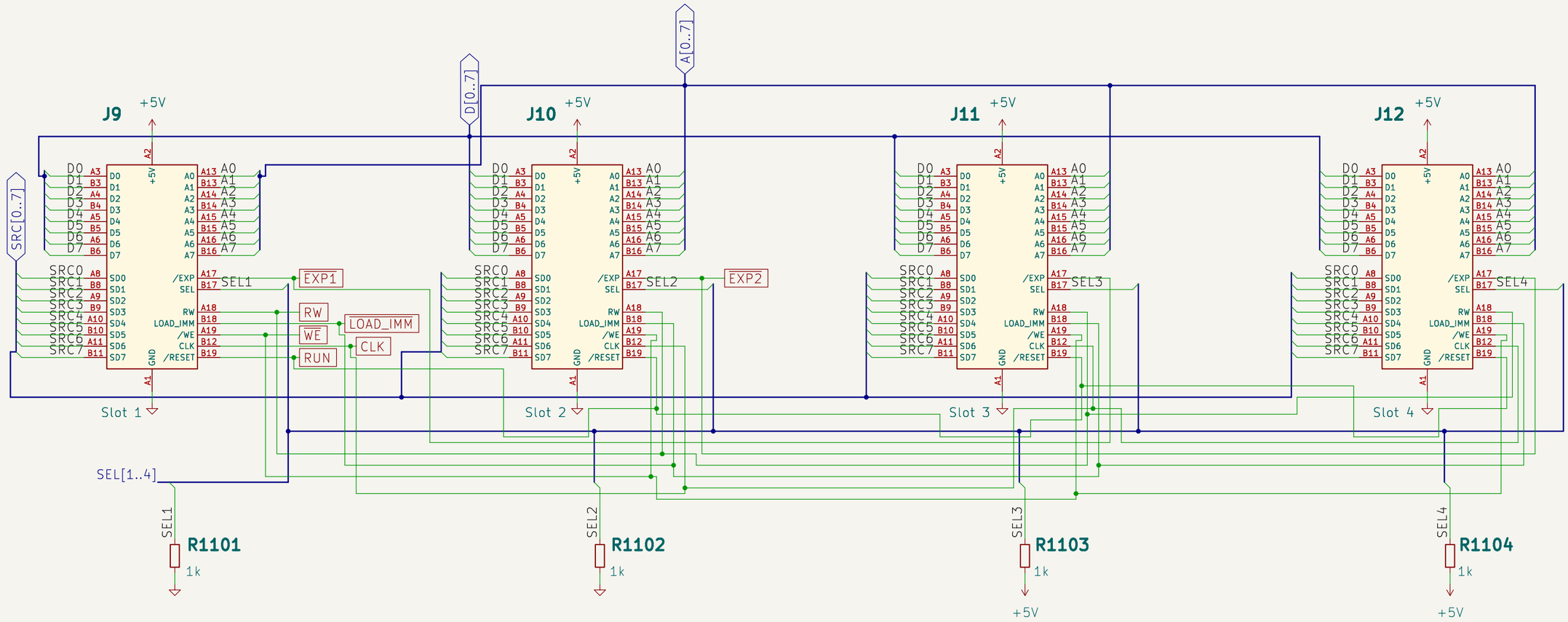
Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

| | |
|---|---|
| Sheet: /Shared port/ | |
| File: shared_port.kicad_sch | |
| ***Title: Placard: Shared port*** | |
| Size: A3 | Date: 2026-10-20 |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 10/11 |

**Rev: 1**

# Expansion slots

Four 2 x 20 card-edge slots, x5100–x53FF, 128 bytes each (EXPANSION.md)




SEL: GND on slots 1 and 2, +5 V on slots 3 and 4 (through 1k): the A7 half the card answers.
/EXP1 goes to slots 1 and 3, /EXP2 to slots 2 and 4. /RESET = RUN (low while the cartridge loads).
SD7:SD0 = SRC7:SRC0, the source drivers' side of the series resistors (the write data).
LOAD_IMM carries the State PLD's /LOAD_IMM (low when the CPU latches the read bus into D).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
**Placard computer**

Sheet: /Expansion slots/
File: expansion.kicad_sch

***Title: Placard: Expansion slots***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 11/11 |